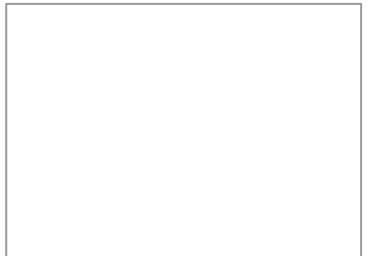


Vanishing gradients
"solved" → Model long sequences ✓
"solved"

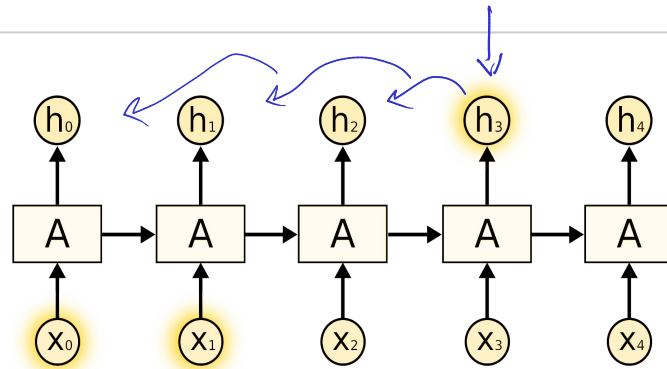
Long Short-term Memory Networks

Building E2E sequence models using deep learning

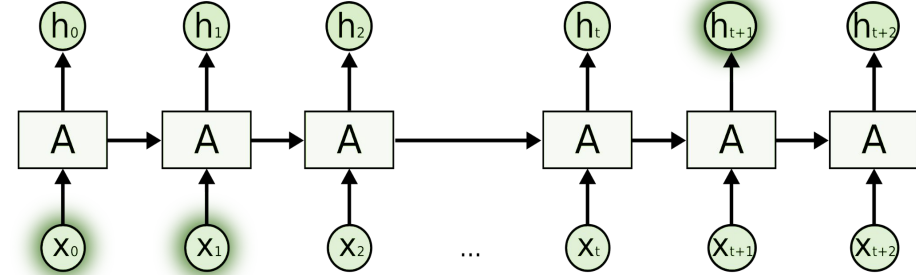
Lakshya Priyadarshi



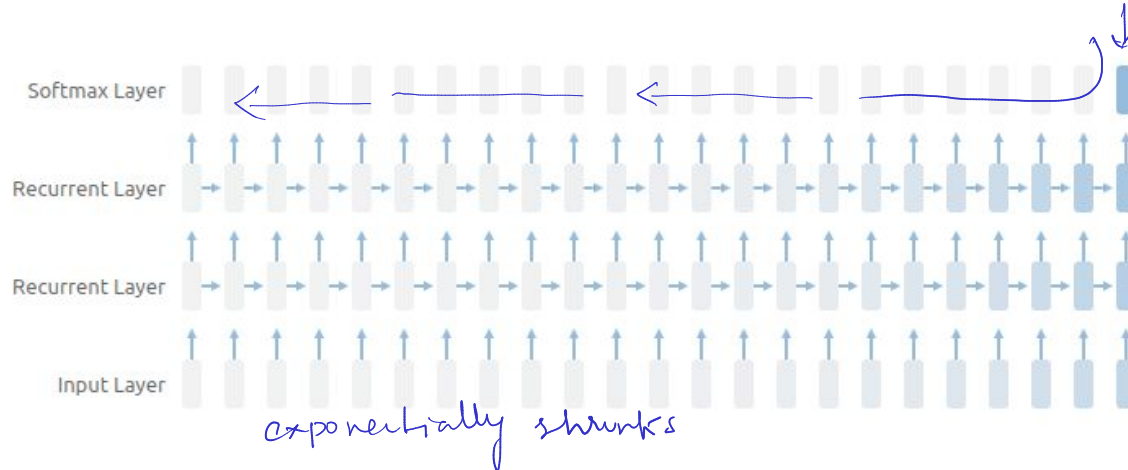
Modeling long term dependencies



"the clouds are in the sky" ✓



"I grew up in France... I speak fluent French"



Why are vanilla recurrent networks not practical?

- RNNs are **not suitable for scalability** of models *↑↑↑↑↑*
- backpropagation training is extremely **slow**
- difficult for training on **long input sequences**
- **due to vanishing gradients,**
- **can not model long-term dependencies**

$$\nabla_{total} = \nabla_{st} + \boxed{\nabla_{lt} \rightarrow 0}$$

dominates